1 Spring–mass models
- Announcements
- IVPs for linear models
- Free undamped motion
- Free damped motion
- Forced motion
- Undamped forcing and resonance
- Practice problems
- Systems of first-order ODEs
- Epidemics as Forced Oscillators

McMaster University

Mathematics and Statistics

$$\int_M d\omega = \int_{\partial M} \omega$$

Mathematics 2C03
Introduction to Differential Equations

Instructor: David Earn

Lecture 24
Spring–mass models: free, damped, and forced motion
Friday 6 November 2026

Announcements

- The TA is Illya Kierkosz (`kierkosi@mcmaster.ca`)
  - Office hours by appointment: e-mail to arrange a meeting.
- Course web site: `https://davidearn.github.io/math2c/`
- The course outline is on the Simple Syllabus web site.
- Avenue to Learn provides secure access to Echo360 live streams and lecture recordings.
- Assignments and class polls are on `childsmath`.
- **e-mail:** PLEASE always include 2C03 in the subject line of any message about this course, and PLEASE indicate what the issue is. For example:
  - `Subject: Math 2C03: confusion about assignment 1`
  - `Subject: Math 2C03: conflict with test window`
- Office hour: Thursdays, 9:30–10:20 am, after class.

Spring–mass models

[Trench §6.1: undamped motion]
[Trench §6.2: damped motion]

IVP for linear model with constant coefficients

Suppose $a_2 \neq 0$ and $g(t)$ is continuous on an interval $\mathcal{I}$ containing t_0. The IVP

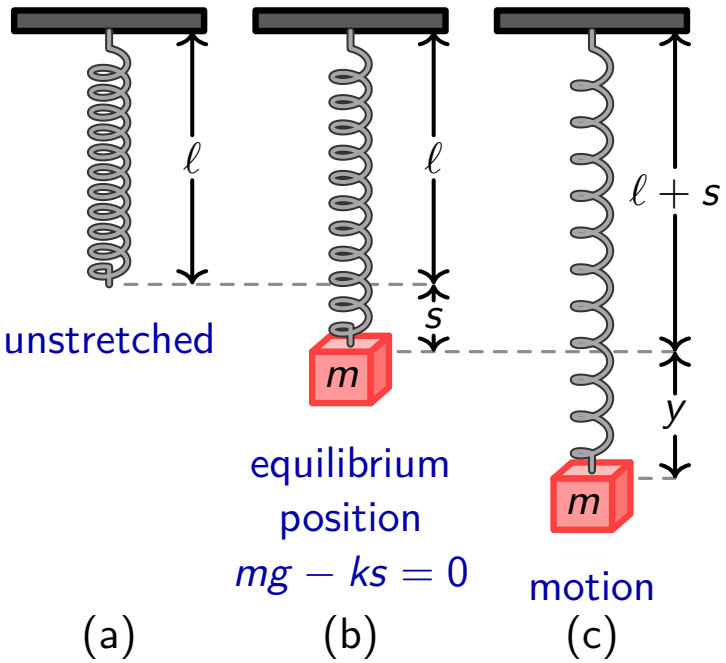
$$a_2 \frac{\mathrm{d}^2 y}{\mathrm{d}t^2} + a_1 \frac{\mathrm{d}y}{\mathrm{d}t} + a_0 y = g(t), \qquad y(t_0) = y_0, \ \dot{y}(t_0) = y_1,$$

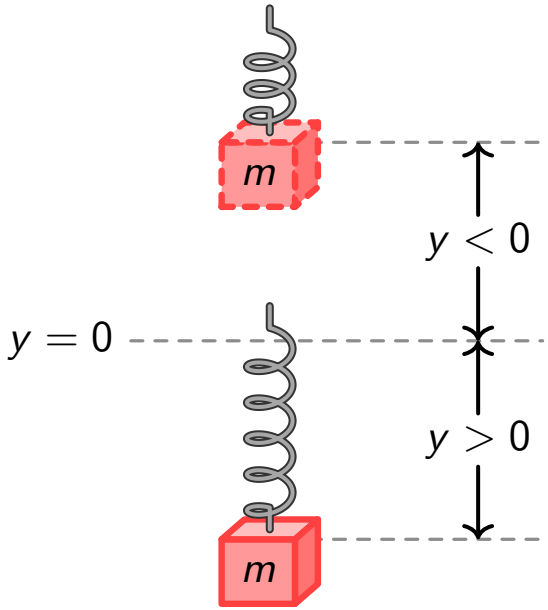
can represent a dynamical system, for which the ***state*** of the system $(y, \dot{y})$ at an initial time t_0 is known, and the state at time t is sought.

Note: To specify the state at time t_0, we need the values of both y and $\dot{y}$ at t_0. Then the future state of the system is determined (according to the existence-uniqueness theorem).

The function $g(t)$ is often called a ***forcing function***, and the solution $y(t)$ is often called the ***output*** or ***response***.

Hooke's Law





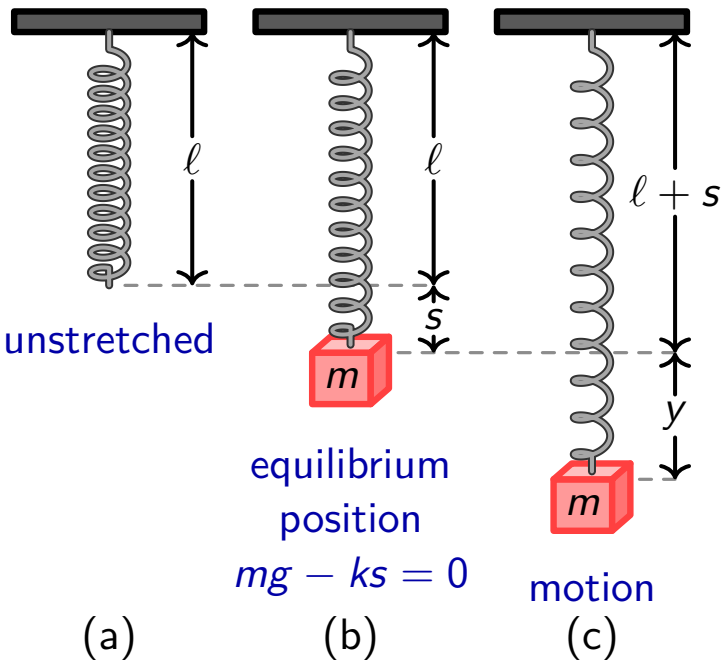
Hooke's Law:

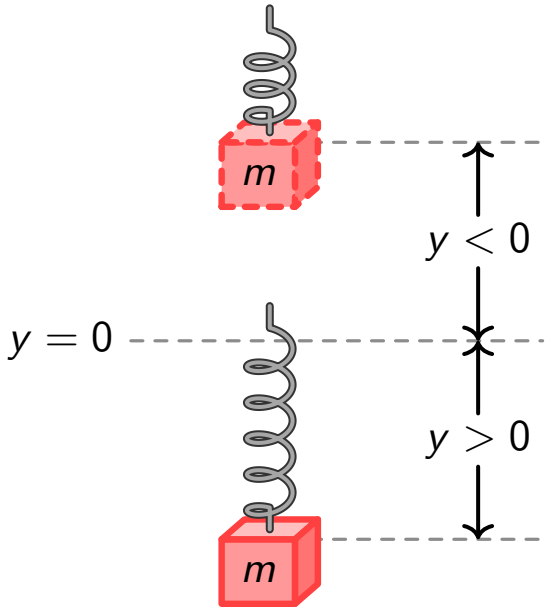
The spring exerts a restoring force F opposite to the direction of stretching, and proportional to the stretch s:

$$F = -ks, \qquad k > 0$$

k is the ***spring constant*** or ***stiffness*** of the spring.

Newton's Second Law





Newton's 2nd Law:

$$m\frac{\mathrm{d}^2y}{\mathrm{d}t^2} = -k(s+y) + mg = -ky + (mg - ks) = -ky$$

$$\Longrightarrow \quad \frac{\mathrm{d}^2y}{\mathrm{d}t^2} + \omega_0^2 y = 0\,, \qquad \omega_0^2 = \frac{k}{m}.$$

Mass, weight, and units

- *Mass* m and *weight* W are different:

$$W = mg, \qquad m = \frac{W}{g}.$$

- At equilibrium, Hooke's law gives

$$ks = W, \qquad k = \frac{W}{s}.$$

- In SI units, use metres, seconds, kilograms, and newtons:

$$g \approx 9.8\,\mathrm{m/s^2}.$$

- If weight is given in pounds and length in feet, use, *e.g.*,

$$\textit{acceleration}: \quad g = 32\,\mathrm{ft/s^2}, \qquad \textit{mass}: \quad \frac{64\,\mathrm{lb}}{32\,\mathrm{ft/s^2}} = 2\,\mathrm{slugs}.$$

  Convert inches to feet before calculating k or initial conditions.

Note: Displacement y is measured positive downward. Trench [§6.1] measures it positive upward, so his displacement has the opposite sign.

Free undamped motion

$$\frac{d^2y}{dt^2} + \omega_0^2 y = 0\,, \qquad \omega_0 = \sqrt{k/m}$$

$\Longrightarrow$ try solution $y = e^{rt}$

$\Longrightarrow$ characteristic equation $r^2 + \omega_0^2 = 0$

$\Longrightarrow$ $r = \pm i\omega_0$

$\Longrightarrow$ general solution $y(t) = c_1 \cos \omega_0 t + c_2 \sin \omega_0 t \qquad (*)$

This ***equation of motion*** (*) describes ***simple harmonic motion*** at ***angular frequency*** ω_0, or ***period*** $T = 2\pi/\omega_0$. The ***frequency*** of motion is $1/T = \omega_0/2\pi$.

Simple harmonic motion is also known as ***free vibration*** (as opposed to ***forced vibration***, which occurs if the equation is non-homogeneous).

Free undamped motion

The equation of motion for simple harmonic motion,

$$y(t) = c_1 \cos \omega_0 t + c_2 \sin \omega_0 t\,, \qquad (*)$$

is more commonly written

$$y(t) = R \cos\left(\omega_0 t - \phi\right), \qquad (**)$$

where the ***amplitude*** R and ***phase*** ϕ are

$$R = \sqrt{c_1^2 + c_2^2} \qquad \text{and} \qquad \begin{aligned} R\cos\phi &= c_1, \\ R\sin\phi &= c_2. \end{aligned}$$

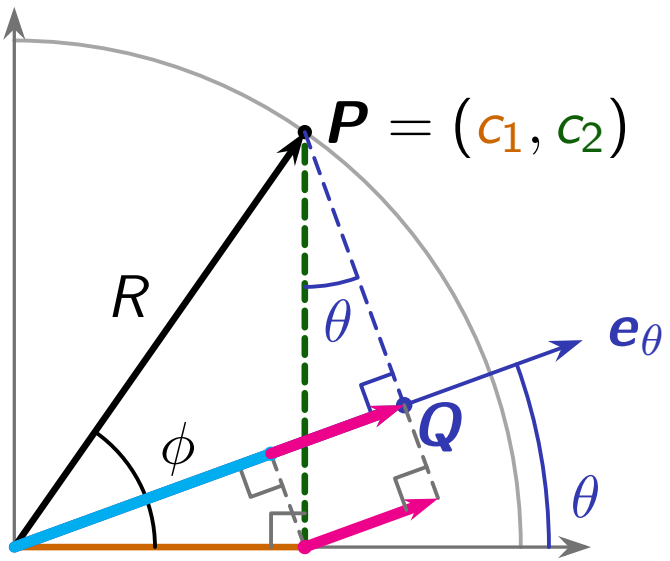
This can be verified using the cosine sum formula:

$$\begin{aligned} R\cos\left(\omega_0 t - \phi\right) &= R\big(\cos\left(\omega_0 t\right)\cos\left(-\phi\right) - \sin\left(\omega_0 t\right)\sin\left(-\phi\right)\big) \\ &= \left(R\cos\phi\right)\cos\omega_0 t + \left(R\sin\phi\right)\sin\omega_0 t \end{aligned}$$

Hence, setting $R\cos\phi = c_1$ and $R\sin\phi = c_2$, we have $R^2\cos^2\phi + R^2\sin^2\phi = R^2 = c_1^2 + c_2^2$ and $\tan\phi = c_2/c_1$ if $c_1 \neq 0$.

Geometry of the amplitude–phase conversion

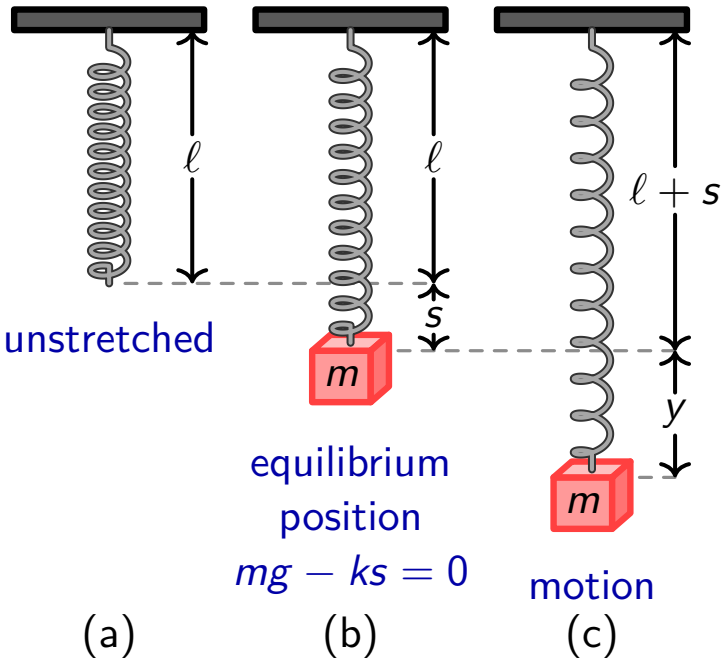
Let $\theta = \omega_0 t$. A picture can help us understand the conversion from the coefficients (c_1 and c_2) of a linear combination of $\cos\theta$ and $\sin\theta$ to amplitude R and phase ϕ: $c_1\cos\theta + c_2\sin\theta = R\cos(\theta - \phi)$.

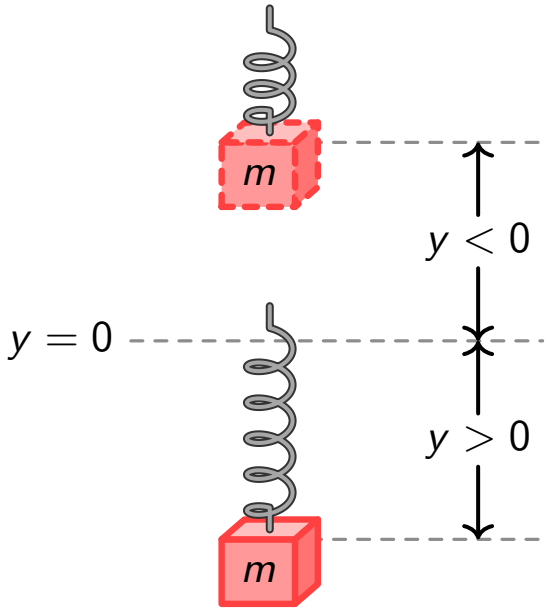


- The vector $\boldsymbol{P} = (c_1, c_2)$ has length R and angle ϕ, so $c_1 = R\cos\phi$ and $c_2 = R\sin\phi$.
- Project $\boldsymbol{P}$ onto the unit direction $\boldsymbol{e}_\theta = (\cos\theta, \sin\theta)$.
- In this first-quadrant picture, the projection has length
$$\begin{aligned}\|\boldsymbol{Q}\| &= R\cos(\phi - \theta)\\ &= R\cos(\theta - \phi).\end{aligned}$$

The horizontal part projects to $c_1\cos\theta$ (angle θ).
The side opposite θ at $\boldsymbol{P}$ has length $c_2\sin\theta$ (hypotenuse c_2).
Opposite rectangle sides are equal, so $\|\boldsymbol{Q}\| = c_1\cos\theta + c_2\sin\theta$.

Free undamped motion





ODE for simple harmonic motion:

$$\frac{\mathrm{d}^2y}{\mathrm{d}t^2} + \omega_0^2 y = 0\,, \quad \omega_0^2 = \frac{k}{m}.$$

Solution (equation of motion):

$$y(t) = R\cos\left(\omega_0 t - \phi\right)$$

Free undamped motion

Example (A brick of mass 2 kg is hanging from a spring)

Suppose the unstretched spring has length $= 0.5\,m$, and the brick is at equilibrium when stretching the spring by 49 cm.

Thus, $m = 2\,\text{kg}$, $\ell = 0.5\,\text{m}$, $s = 0.49\,\text{m}$.

At equilibrium, we know $mg - ks = 0$ so
$k = mg/s = (2\,\text{kg})(9.8\,\text{m/s}^2)/(0.49\,\text{m}) = 40\,\text{kg/s}^2$.

$$\omega_0 = \sqrt{k/m} = \sqrt{(40\,\text{kg/s}^2)/(2\,\text{kg})} = \sqrt{20}\,\text{s}^{-1} \approx 4.47\,\text{s}^{-1}$$

$$T = 2\pi/\omega_0 \approx 1.40\,\text{s}$$

Continued...

Free undamped motion

Example (A brick of mass 2 kg is hanging from a spring)

Suppose the brick is tapped, so it is initially moving down at a speed of 10 cm/s. What is its equation of motion?

$y(t) = R\cos(\omega_0 t - \phi) \implies \dot{y}(t) = -R\,\omega_0 \sin(\omega_0 t - \phi)$

Initial conditions: $y(0) = 0$, $\dot{y}(0) = 0.1\,\text{m/s}$.

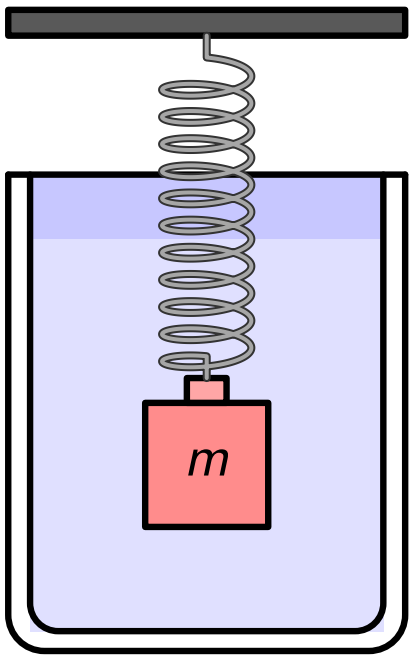
$y(0) = 0 = R\cos(-\phi) = R\cos\phi \implies R = 0$ or $\phi = \frac{\pi}{2} + n\pi$, $n \in \mathbb{Z}$.

$\dot{y}(0) = 0.1\,\text{m/s} = R\,\omega_0 \sin\phi \implies \sin\phi > 0 \implies n$ even $\implies$ $\sin\phi = \sin\left(\frac{\pi}{2}\right) = 1 \implies R\,\omega_0 = \dot{y}(0)$

$\implies R = (0.1\,\text{m/s})/(4.47\,\text{s}^{-1}) = 0.1/4.47\,\text{m} \approx 0.0224\,\text{m}$

$\therefore R = 2.24\,\text{cm} \implies \boxed{y(t) \approx (2.24\,\text{cm})\cos((4.47\,\text{s}^{-1})t - 1.57)}$

Free damped motion



Resistance *damps* oscillations

$$m\frac{\mathrm{d}^2y}{\mathrm{d}t^2} = -ky - c\frac{\mathrm{d}y}{\mathrm{d}t}$$

$c > 0$ is the *damping constant*.

Standard form:

$$\frac{\mathrm{d}^2y}{\mathrm{d}t^2} + \frac{c}{m}\frac{\mathrm{d}y}{\mathrm{d}t} + \frac{k}{m}y = 0$$

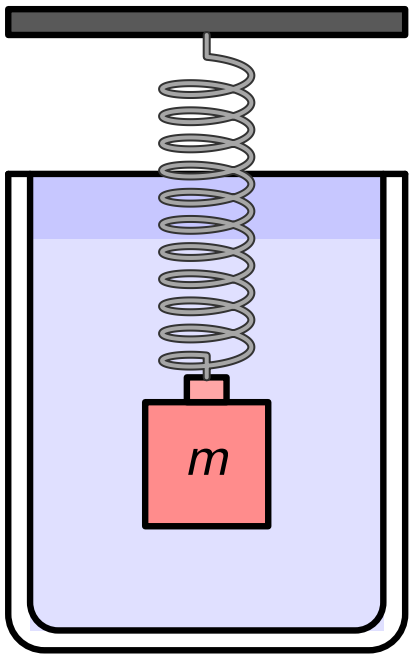
Standard form with convenient parameters:

$$\frac{\mathrm{d}^2y}{\mathrm{d}t^2} + 2\lambda\frac{\mathrm{d}y}{\mathrm{d}t} + \omega_0^2 y = 0$$

$$2\lambda = c/m, \quad \omega_0 = \sqrt{k/m}$$

Free damped motion

$$\frac{d^2y}{dt^2} + 2\lambda\frac{dy}{dt} + \omega_0^2\, y = 0$$



characteristic equation: $r^2 + 2\lambda r + \omega_0^2 = 0$

$$\Longrightarrow \quad r = -\lambda \pm \sqrt{\lambda^2 - \omega_0^2}$$

Case (i): $\lambda^2 - \omega_0^2 > 0$
(***overdamped***)

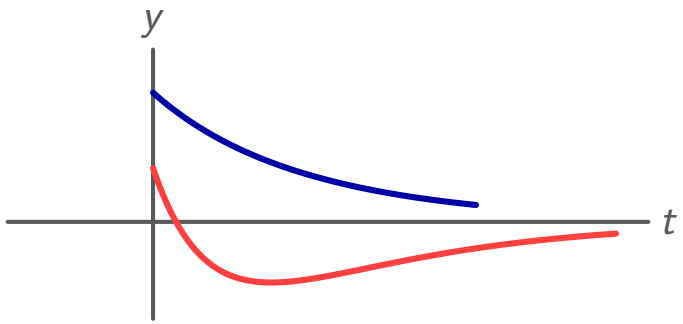
Case (ii): $\lambda^2 - \omega_0^2 = 0$
(***critically damped***)

Case (iii): $\lambda^2 - \omega_0^2 < 0$
(***underdamped***)

Free damped motion

Case (i): $\lambda^2 - \omega_0^2 > 0$
(***overdamped***)

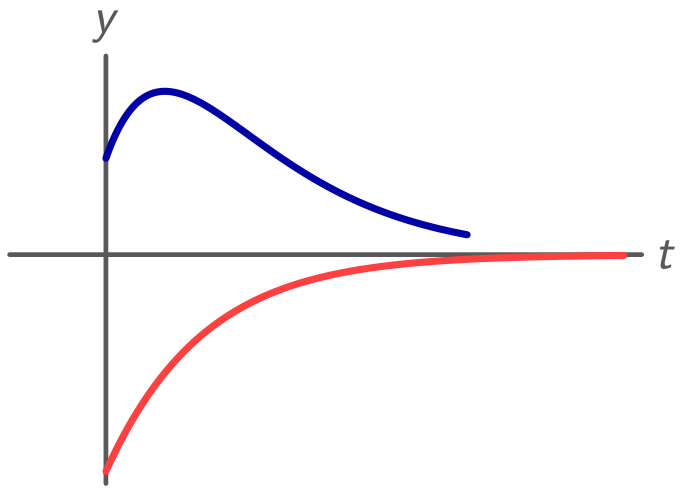
$$y(t) = e^{-\lambda t}\left(c_1 e^{\sqrt{\lambda^2-\omega_0^2}\,t} + c_2 e^{-\sqrt{\lambda^2-\omega_0^2}\,t}\right)$$



Note: $\lim_{t\to\infty} y(t) = 0.$

Case (ii): $\lambda^2 - \omega_0^2 = 0$
(***critically damped***)

$$y(t) = e^{-\lambda t}(c_1 + c_2\, t)$$



Note: $\lim_{t\to\infty} y(t) = 0.$

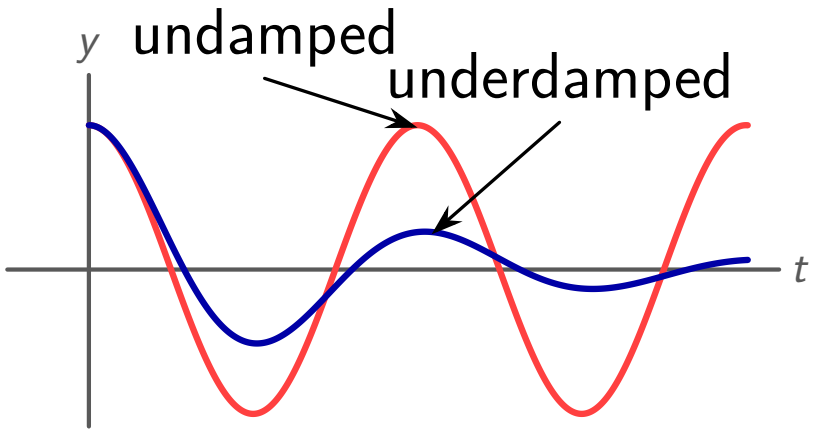
Free damped motion

Case (iii): $\lambda^2 - \omega_0^2 < 0$ (***underdamped***)

Let $\omega_1 = \sqrt{\omega_0^2 - \lambda^2}$.

$$y(t) = e^{-\lambda t}\left(c_1 \cos(\omega_1 t) + c_2 \sin(\omega_1 t)\right)$$

$$= Re^{-\lambda t} \cos\left((\omega_1 t) - \phi\right)$$



Note: $\lim_{t\to\infty} y(t) = 0$.

- Recall $2\lambda = c/m > 0$. For any $\lambda > 0$, *i.e.*, any amount of damping, $y(t) \to 0$ as $t \to \infty$, regardless of value of ω_0.
- Hence, any amount of damping makes $y = 0$ a *stable equilibrium*: initial displacements from $y = 0$ always decay (they "*damp out*").

Equilibrium crossings and extreme displacement

Example ($y'' + 3y' + 2y = 0, \quad y(0) = -1, \quad y'(0) = 5$)

The characteristic equation is

$$(r+1)(r+2) = 0 \quad \Longrightarrow \quad y(t) = c_1 e^{-t} + c_2 e^{-2t}.$$

The initial conditions give

$$c_1 + c_2 = -1, \quad -c_1 - 2c_2 = 5 \quad \Longrightarrow \quad c_1 = 3, \quad c_2 = -4.$$

Thus $y(t) = 3e^{-t} - 4e^{-2t}$.

- *Equilibrium crossing:*

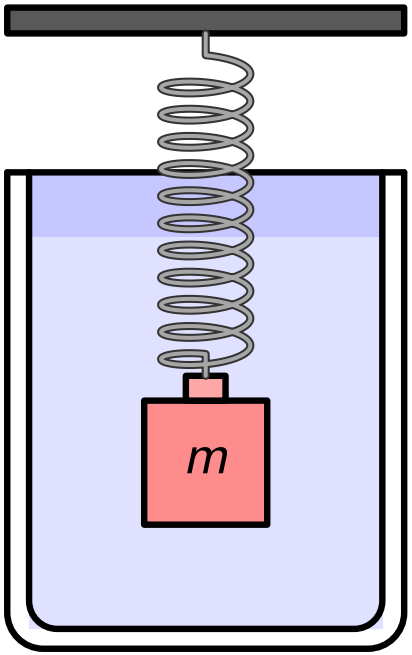
$$y(t) = 0 \quad \Longrightarrow \quad 3e^{t} - 4 = 0 \quad \Longrightarrow \quad \boxed{t = \ln(4/3)}$$

- *Extreme displacement:*

$$y'(t) = -3e^{-t} + 8e^{-2t} = 0 \quad \Longrightarrow \quad \boxed{t = \ln(8/3)}$$

This is a maximum: y' changes from positive to negative.

Forced damped motion



Suppose we apply an ***external force*** $F(t)$ to the brick, *e.g.*, we could push or pull the brick in some way.

$$m\frac{\mathrm{d}^2y}{\mathrm{d}t^2} = -k\,y - c\frac{\mathrm{d}y}{\mathrm{d}t} + F(t)$$

Standard form:

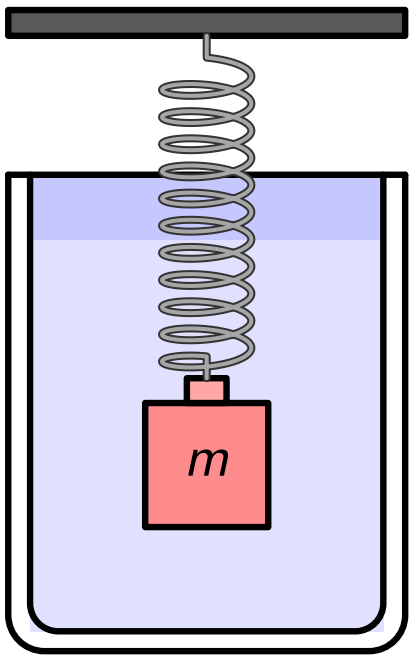
$$\frac{\mathrm{d}^2y}{\mathrm{d}t^2} + \frac{c}{m}\frac{\mathrm{d}y}{\mathrm{d}t} + \frac{k}{m}\,y = \frac{F(t)}{m}$$

Standard form with convenient parameters:

$$\frac{\mathrm{d}^2y}{\mathrm{d}t^2} + 2\lambda\frac{\mathrm{d}y}{\mathrm{d}t} + \omega_0^2\,y = \frac{F(t)}{m}$$

$$2\lambda = c/m, \quad \omega_0 = \sqrt{k/m}$$

Periodically forced oscillator



Suppose the system is underdamped, so it undergoes *damped oscillations*.

Now suppose we attach a motor that pushes or pulls the brick periodically, such that the external force is $F(t) = F_0 \sin \omega t$.

Dividing by m, we have

$$\frac{\mathrm{d}^2 y}{\mathrm{d}t^2} + 2\lambda \frac{\mathrm{d}y}{\mathrm{d}t} + \omega_0^2 \, y = \frac{F_0}{m} \sin \omega t$$

What types of motion are possible for this periodically forced oscillator?

Periodically forced oscillator

$$L[y] = \frac{\mathrm{d}^2 y}{\mathrm{d}t^2} + 2\lambda \frac{\mathrm{d}y}{\mathrm{d}t} + \omega_0^2 y = \frac{F_0}{m} \sin \omega t$$

The general solution of this equation can be written $y(t) = y_c(t) + y_p(t)$, where:

- $y_c(t)$ is the general solution to $L[y] = 0$, which we know decays because it contains a factor $e^{-\lambda t}$;
- $y_p(t)$ is the particular solution with no decaying component;
  - any other particular solution contains terms from $y_c(t)$, which decay;
  - find $y_p(t)$ via undetermined coefficients;

$y_c(t)$ specifies the ***transient dynamics***, whereas $y_p(t)$ specifies the ***asymptotic dynamics*** (what remains after the transient behaviour has damped out).

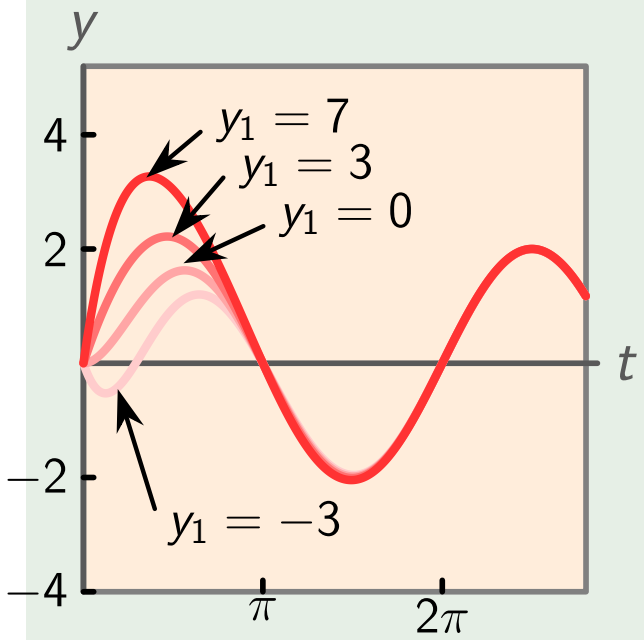
Note: Trench [after Eq. (6.2.17)] uses the term ***steady state component*** for the asymptotic dynamics.

Periodically forced oscillator

Example (Transient and asymptotic dynamics)

IVP: $$\frac{d^2y}{dt^2}+2\frac{dy}{dt}+2y = 4\cos t+2\sin t\,, \quad y(0)=0,\ y'(0)=y_1.$$

Solution: $$y(t) = \underbrace{(y_1-2)}_{\text{constant}}\underbrace{e^{-t}\sin t}_{\text{transient}} + \underbrace{2\sin t}_{\text{asymptotic}}$$



- all solutions start at equilibrium ($y = 0$);
- different solutions have different initial velocities (y_1);
- transient dynamics damp out quickly;
- all solutions quickly converge onto the asymptotic dynamics (what Trench calls the "steady state component").

Periodically forced oscillator

Example (Undamped forced motion)

IVP: $$\frac{\mathrm{d}^2y}{\mathrm{d}t^2} + \omega_0^2 y = \frac{F_0}{m}\sin\omega t\,, \quad y(0) = 0,\ y'(0) = 0, \quad \omega \neq \omega_0.$$

Complementary function:

$$y_c(t) \;=\; R\cos\left(\omega_0 t - \phi\right), \qquad R \geq 0,\ \phi \in [0, 2\pi).$$

Particular solution via undetermined coefficients:

$$y_p(t) \;=\; \frac{F_0}{m(\omega_0^2 - \omega^2)}\sin\omega t\,.$$

General solution:

$$y(t) \;=\; R\cos\left(\omega_0 t - \phi\right) + \frac{F_0}{m(\omega_0^2 - \omega^2)}\sin\omega t\,.$$

Periodically forced oscillator

Example (Undamped forced motion)

$$y(t) = R\cos(\omega_0 t - \phi) + \frac{F_0}{m(\omega_0^2 - \omega^2)}\sin\omega t$$

$$y'(t) = -R\,\omega_0 \sin(\omega_0 t - \phi) + \frac{\omega F_0}{m(\omega_0^2 - \omega^2)}\cos\omega t\,.$$

The initial conditions give

$$R\cos\phi = 0, \qquad R\omega_0 \sin\phi + \frac{\omega F_0}{m(\omega_0^2 - \omega^2)} = 0$$

$$\text{Hence} \quad R\cos\phi = 0, \qquad R\sin\phi = -\frac{\omega F_0}{m\omega_0(\omega_0^2 - \omega^2)}.$$

Using amplitude–phase form to expand the complementary term,

$$y(t) = R\cos\phi\cos\omega_0 t + R\sin\phi\sin\omega_0 t + \frac{F_0}{m(\omega_0^2 - \omega^2)}\sin\omega t$$

$$= \frac{F_0}{m(\omega_0^2 - \omega^2)}\left(-\frac{\omega}{\omega_0}\sin\omega_0 t + \sin\omega t\right).$$

We assumed $\omega \neq \omega_0$. *What happens if $\omega = \omega_0$?*

Periodically forced oscillator

Example (Undamped forced motion)

To solve, either start over with $\omega = \omega_0$ or consider the limit $\omega \to \omega_0$:

$$\lim_{\omega\to\omega_0} y(t) = \lim_{\omega\to\omega_0} \left[\frac{F_0}{m(\omega_0^2 - \omega^2)} \left(-\frac{\omega}{\omega_0} \sin(\omega_0 t) + \sin \omega t \right) \right]$$

This limit has form $\frac{0}{0}$. *Use L'Hôpital's rule ...*

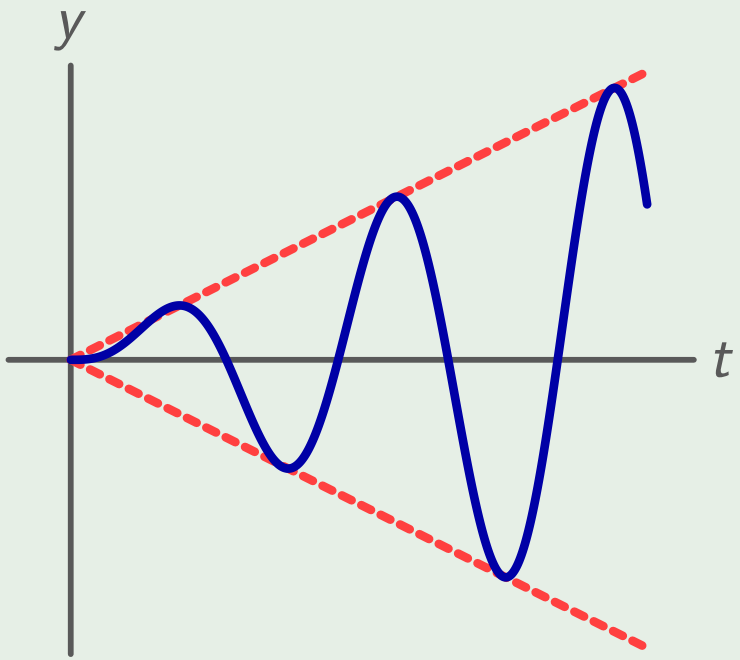
$$y(t) = \frac{F_0}{2m\omega_0^2} \sin \omega_0 t - \frac{F_0}{2m\omega_0} t \cos \omega_0 t$$

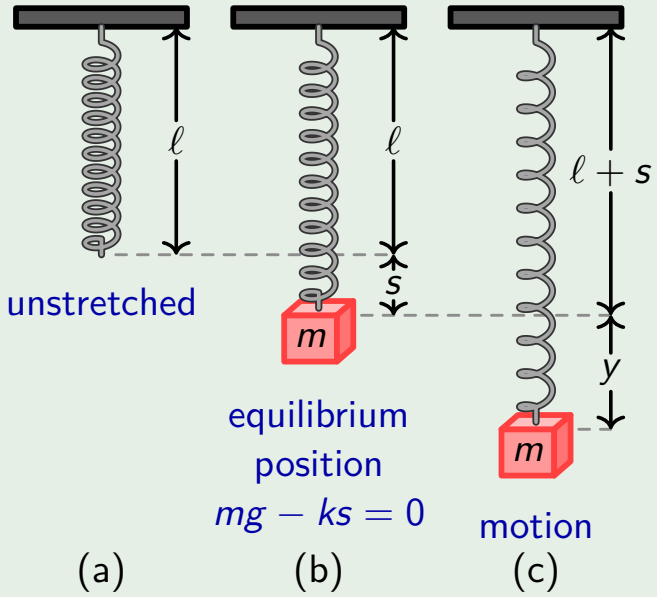
Amplitude $R = \sqrt{\left(\frac{F_0}{2m\omega_0^2}\right)^2 + \left(\frac{F_0}{2m\omega_0} t\right)^2} = \frac{F_0}{2m\omega_0^2}\sqrt{1 + (\omega_0 t)^2}$

Amplitude grows with time t: ***resonance***.

Periodically forced oscillator

Example (Undamped forced motion)





Does this make sense physically for arbitrarily large t?

- Please do Spring resonance poll.

Can the amplitude exceed the spring length ℓ? Can we have $|y| > \ell + s$? For large enough amplitudes of oscillation, modelling the system with a linear ODE is inadequate. We must consider nonlinearity in the restoring force, or some other nonlinear effect(s).

Poll

- Go to
  https://www.childsmath.ca/childsa/forms/main_login.php
- Click on Math 2C03
- Click on Take Class Poll
- Fill in poll **Spring resonance**
- Submit.

A decaying force need not leave a motionless mass

Example ($y'' + 4y = 10e^{-t}, \quad y(0) = y'(0) = 0$)

The complementary solution is $y_c = c_1 \cos 2t + c_2 \sin 2t$. Try $y_p = Ae^{-t}$:

$$y_p'' + 4y_p = 5Ae^{-t} = 10e^{-t} \quad \Longrightarrow \quad A = 2.$$

Thus $y = c_1 \cos 2t + c_2 \sin 2t + 2e^{-t}$. The initial conditions give

$$c_1 + 2 = 0, \qquad 2c_2 - 2 = 0, \quad \Longrightarrow \quad c_1 = -2, \quad c_2 = 1.$$

Therefore

$$\boxed{y(t) = -2\cos 2t + \sin 2t + 2e^{-t}}$$

As $t \to \infty$, the last term decays, leaving an oscillation with amplitude $R = \sqrt{(-2)^2 + 1^2} = \sqrt{5}$. There is no damping to remove this remaining motion.

Trench practice problems

- [Trench §6.1: undamped spring motion]
  - Free motion and initial conditions: 1, 4, 6, 9.
  - Forced motion and resonance: 11.
- [Trench §6.2: damped spring motion]
  - Free damped motion: 1, 3, 7, 9.
  - Forced motion and asymptotic dynamics (Trench's "steady state component"): 17, 20.
  - Equilibrium crossings: 22, 23.

Remember: Trench [§6.1] measures displacement positive upward. Translate the initial conditions and the sign of any external force consistently if using our downward-positive convention.

Systems of first-order ODEs

[Trench §10.1: Intro to systems of ODEs]

The oscillator as two first-order ODEs

Our spring–mass equation is

$$m\frac{\mathrm{d}^2y}{\mathrm{d}t^2} + c\frac{\mathrm{d}y}{\mathrm{d}t} + ky = F(t), \qquad m > 0.$$

- Introduce the velocity as a new unknown function: $v(t) = \frac{\mathrm{d}y}{\mathrm{d}t}$.
- Then $\frac{\mathrm{d}v}{\mathrm{d}t} = \frac{\mathrm{d}^2y}{\mathrm{d}t^2}$. The resulting ***first-order system*** is

$$\boxed{\begin{aligned} \frac{\mathrm{d}y}{\mathrm{d}t} &= v, \\ \frac{\mathrm{d}v}{\mathrm{d}t} &= \frac{1}{m}F(t) - \frac{c}{m}v - \frac{k}{m}y \end{aligned}}$$

The initial conditions become $y(t_0) = y_0$ and $v(t_0) = y_1$. We have two first-order equations and two initial conditions, instead of one second-order equation and two initial conditions.

[Trench §10.1, Examples 10.1.4 and 10.1.7]

The same oscillator, written as a system

For simple harmonic motion, $c = 0$ and $F(t) = 0$. Our system becomes

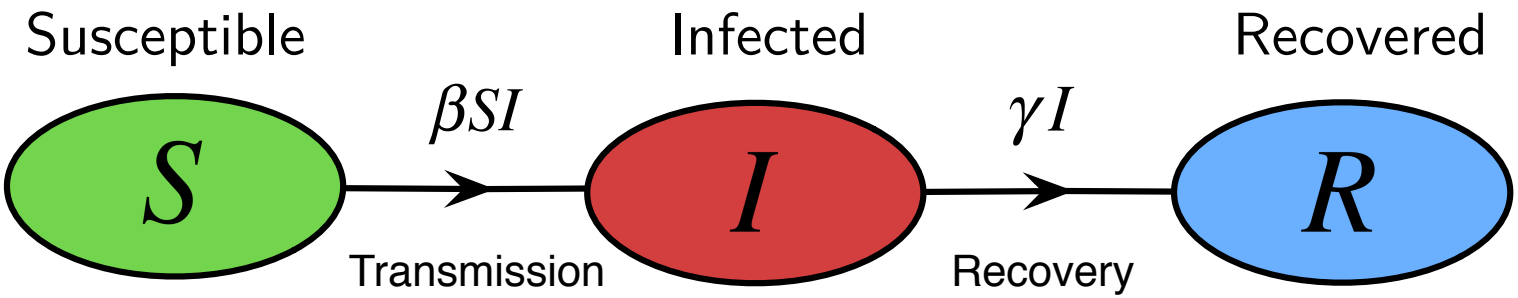
$$\boxed{\begin{aligned} \frac{dy}{dt} &= v, \\ \frac{dv}{dt} &= -\omega_0^2 y, \end{aligned}} \qquad \omega_0^2 = \frac{k}{m}.$$

- The first equation says that v is the velocity. The second specifies the acceleration.
- Conversely, differentiating $\frac{dy}{dt} = v$ and using the second equation gives $\frac{d^2y}{dt^2} + \omega_0^2 y = 0$.
- A system can also describe several quantities in their own right, rather than one quantity and its derivatives.
- In epidemic models, we will track susceptible, infected and recovered populations with a first-order ODE for each.

Nonlinear oscillators and Epidemics

The SIR model: ODEs that describe epidemics

Divide population into compartments:



Express idea with a ***system of three ODEs***:

$$\frac{dS}{dt} = -\beta SI, \qquad \frac{dI}{dt} = \beta SI - \gamma I, \qquad \frac{dR}{dt} = \gamma I$$

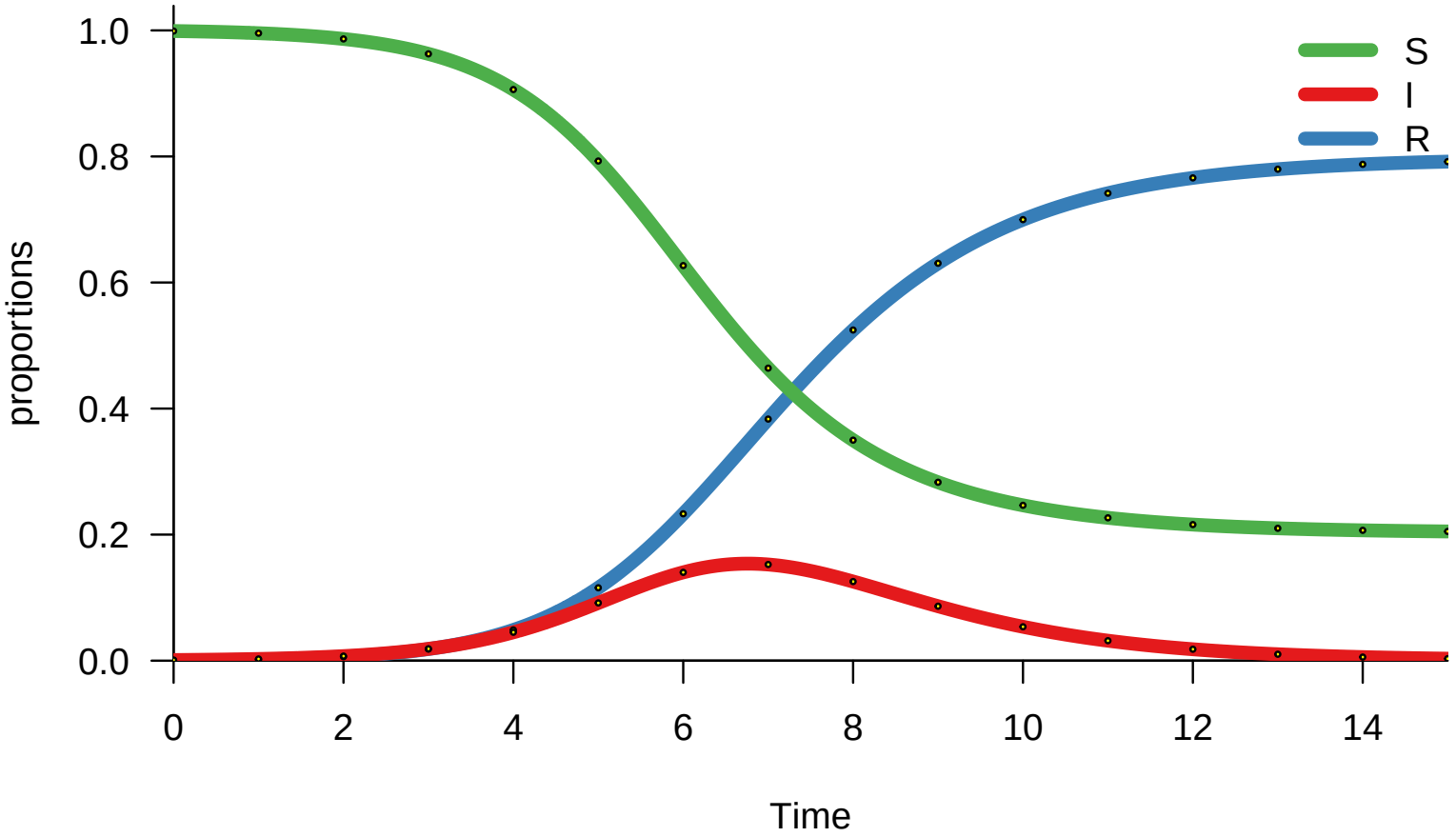
State variables are proportions: $S + I + R = 1$.

Parameters: $\beta =$ ***transmission rate***, $\gamma =$ ***recovery rate***

$$\mathcal{R}_0 = \frac{\beta}{\gamma} = \textbf{\textit{basic reproduction number}}$$

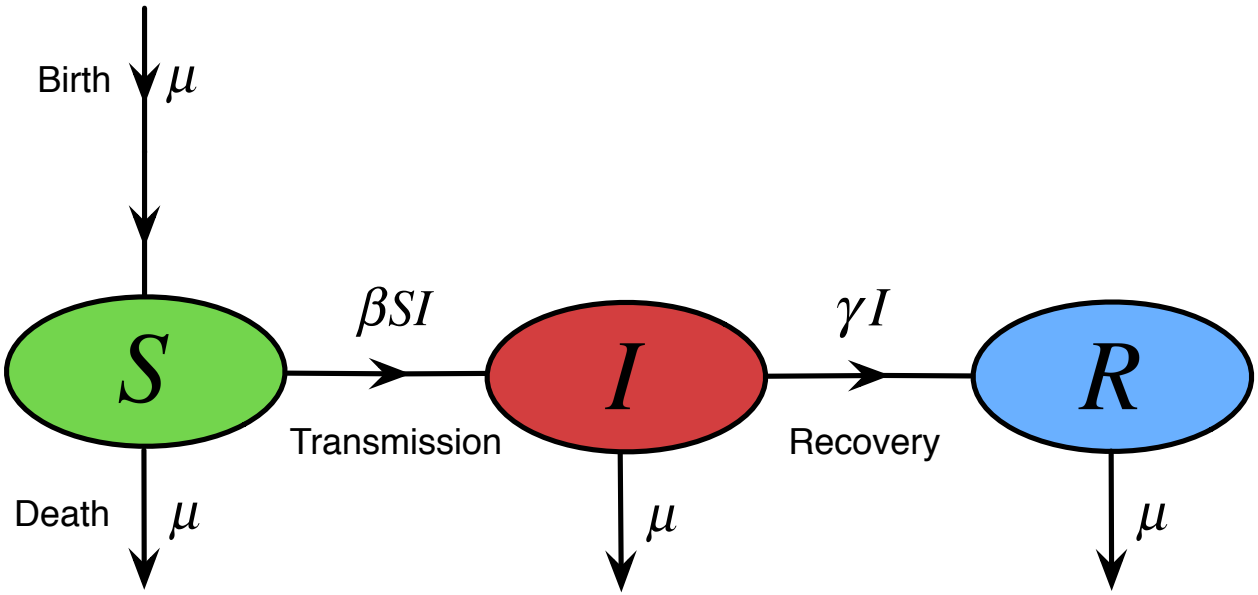
Predicting epidemics with ODEs

Solution is a set of three functions: $S(t)$, $I(t)$, $R(t)$



Predicting epidemics with ODEs

Include ***vital dynamics*** (births and deaths):



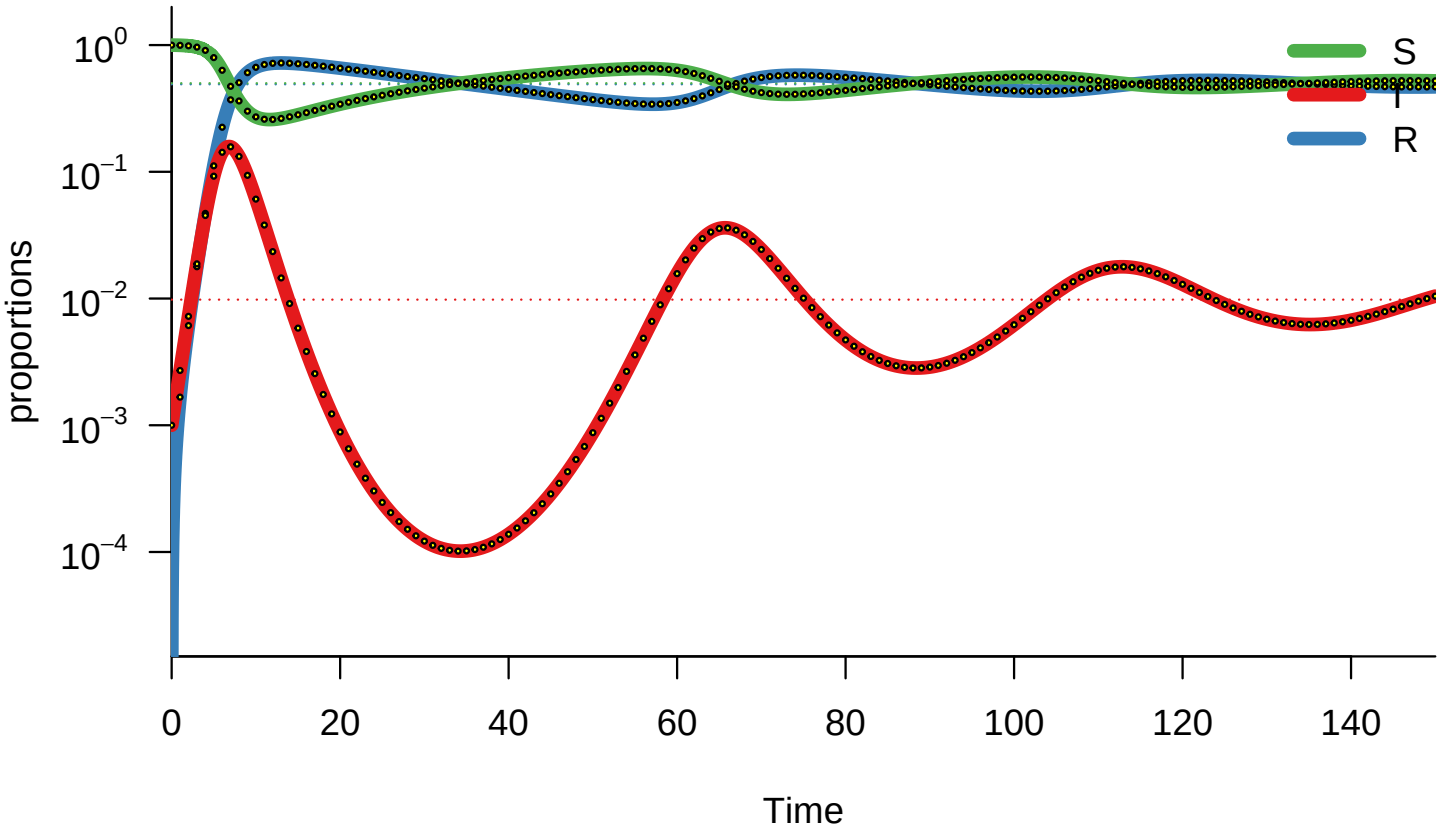
Revised ***system of three ODEs***:

$$\frac{dS}{dt} = \mu - \beta SI - \mu S, \qquad \frac{dI}{dt} = \beta SI - \gamma I - \mu I, \qquad \frac{dR}{dt} = \gamma I - \mu R$$

Additional parameter: $\mu =$ ***birth/death rate*** *(per capita)*

Predicting epidemics with ODEs

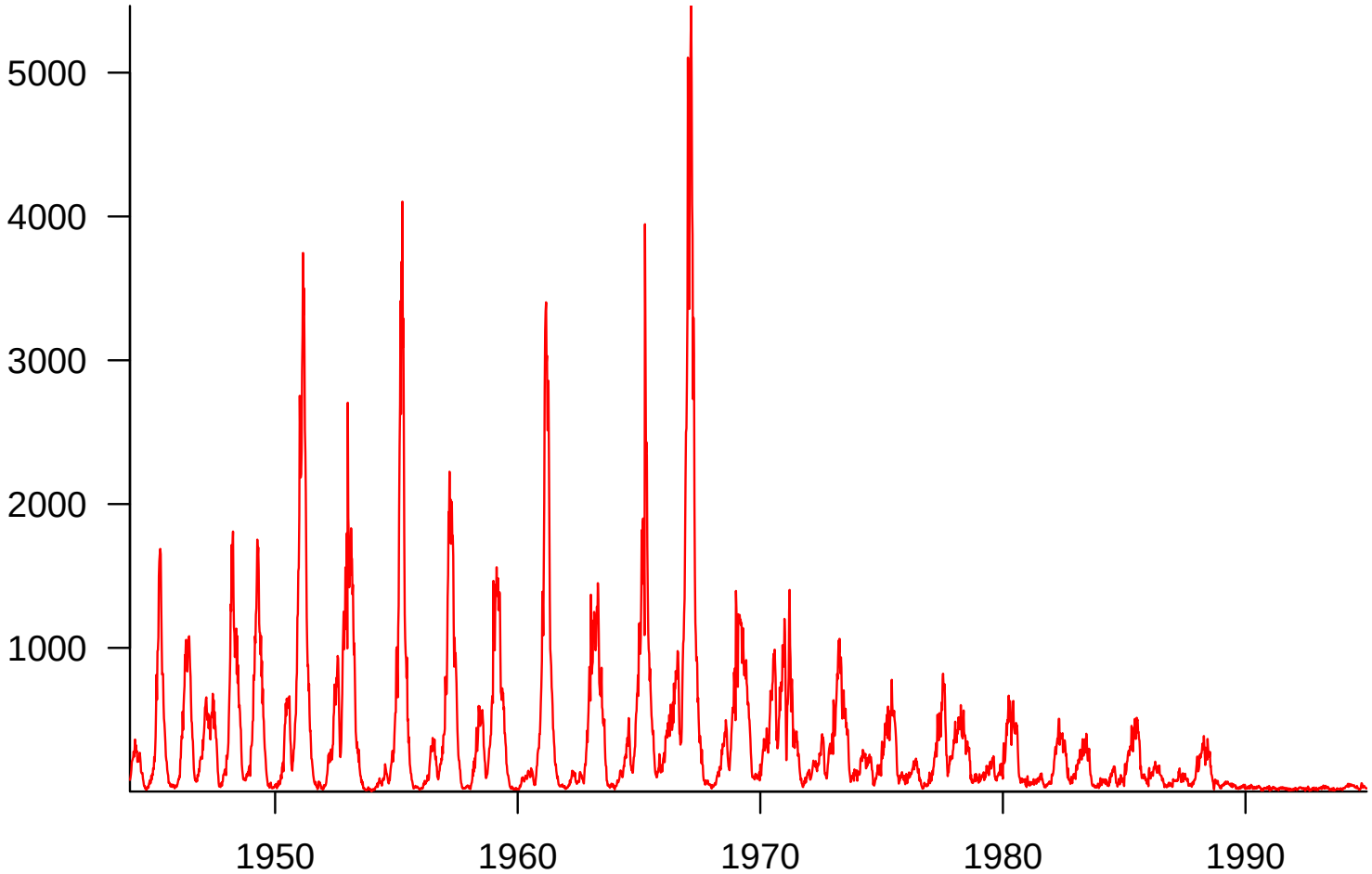
Solution is a set of three functions: $S(t)$, $I(t)$, $R(t)$



The solution displays *damped oscillations* onto an *equilibrium*.
At equilibrium, there is *constant prevalence of disease*.
Do real epidemics damp out in this way?

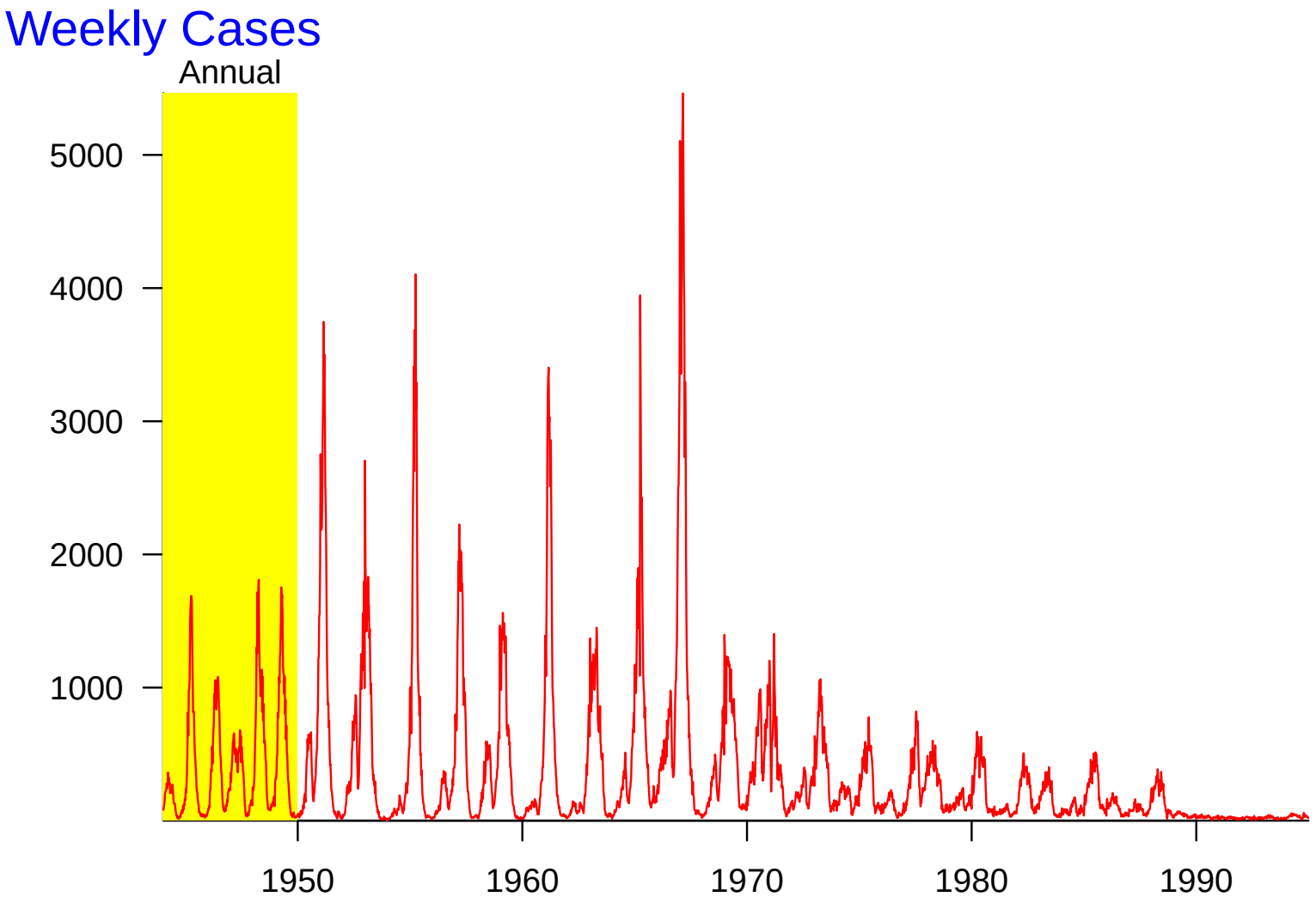
Measles incidence in London, England, 1945–1995

Weekly Cases



Earn, Rohani, Bolker, Grenfell, 2000, *Science* **287**, 667–670

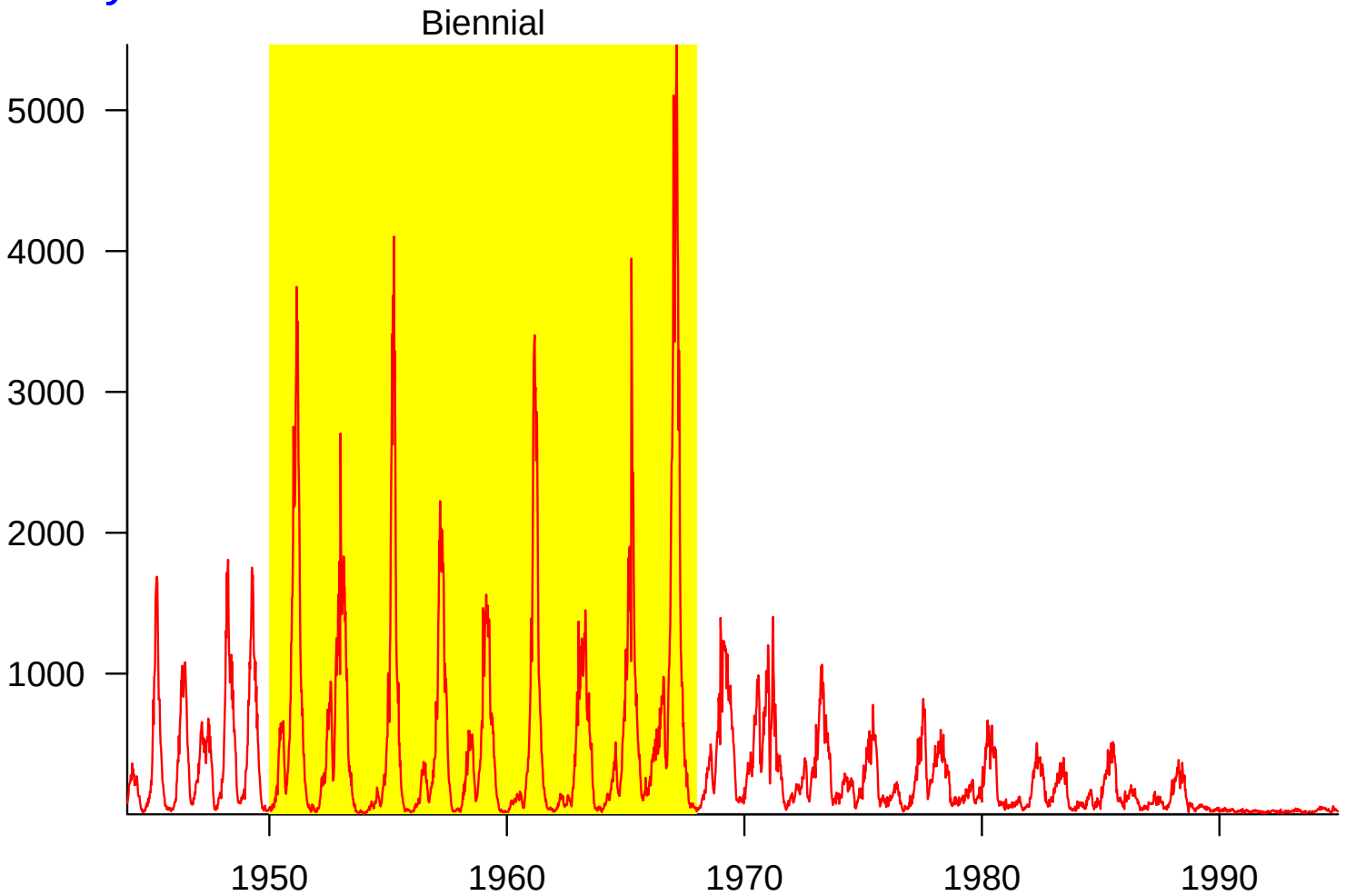
Measles incidence in London, England, 1945–1995



Earn, Rohani, Bolker, Grenfell, 2000, *Science* **287**, 667–670

Measles incidence in London, England, 1945–1995

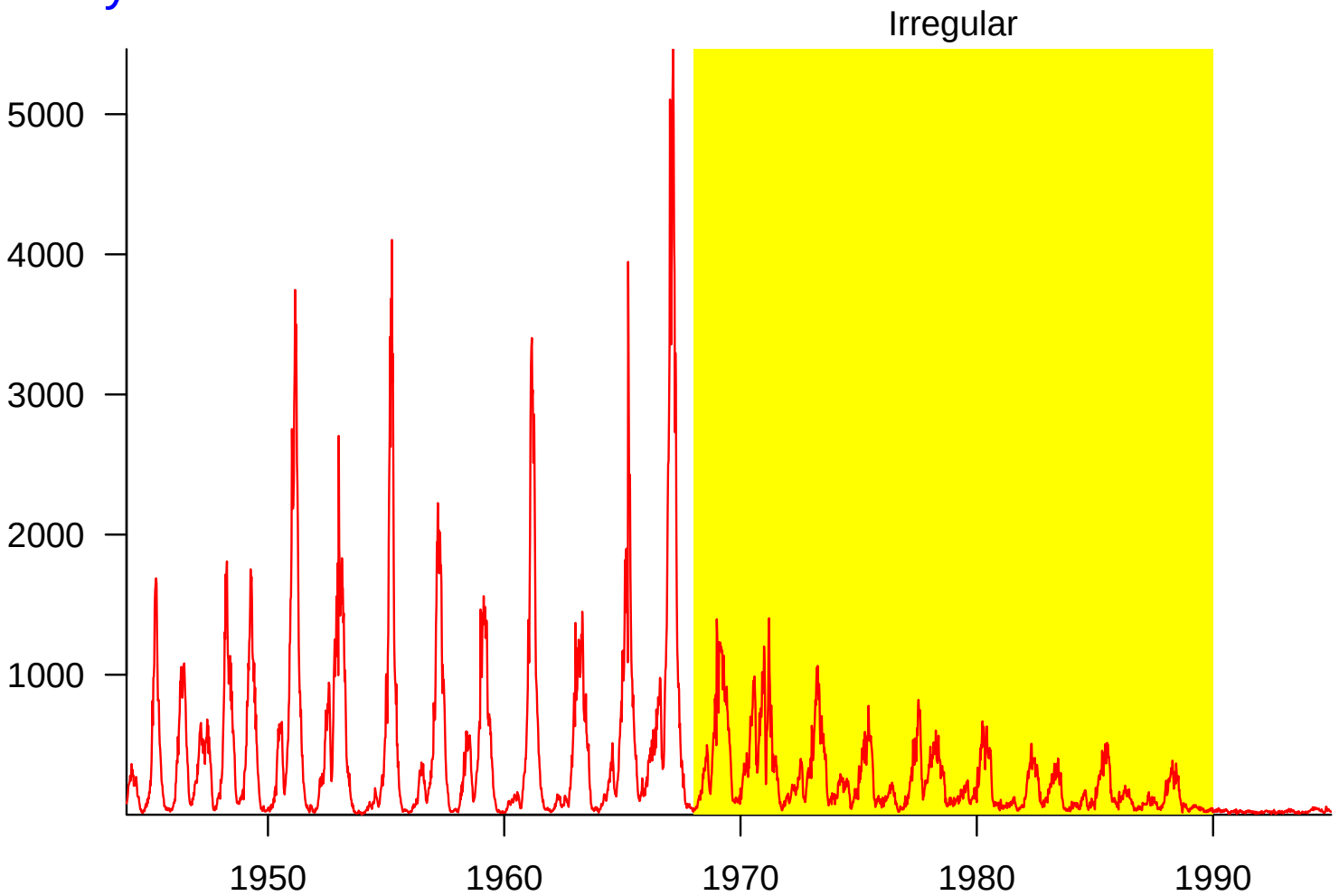
Weekly Cases



Earn, Rohani, Bolker, Grenfell, 2000, *Science* **287**, 667–670

Measles incidence in London, England, 1945–1995

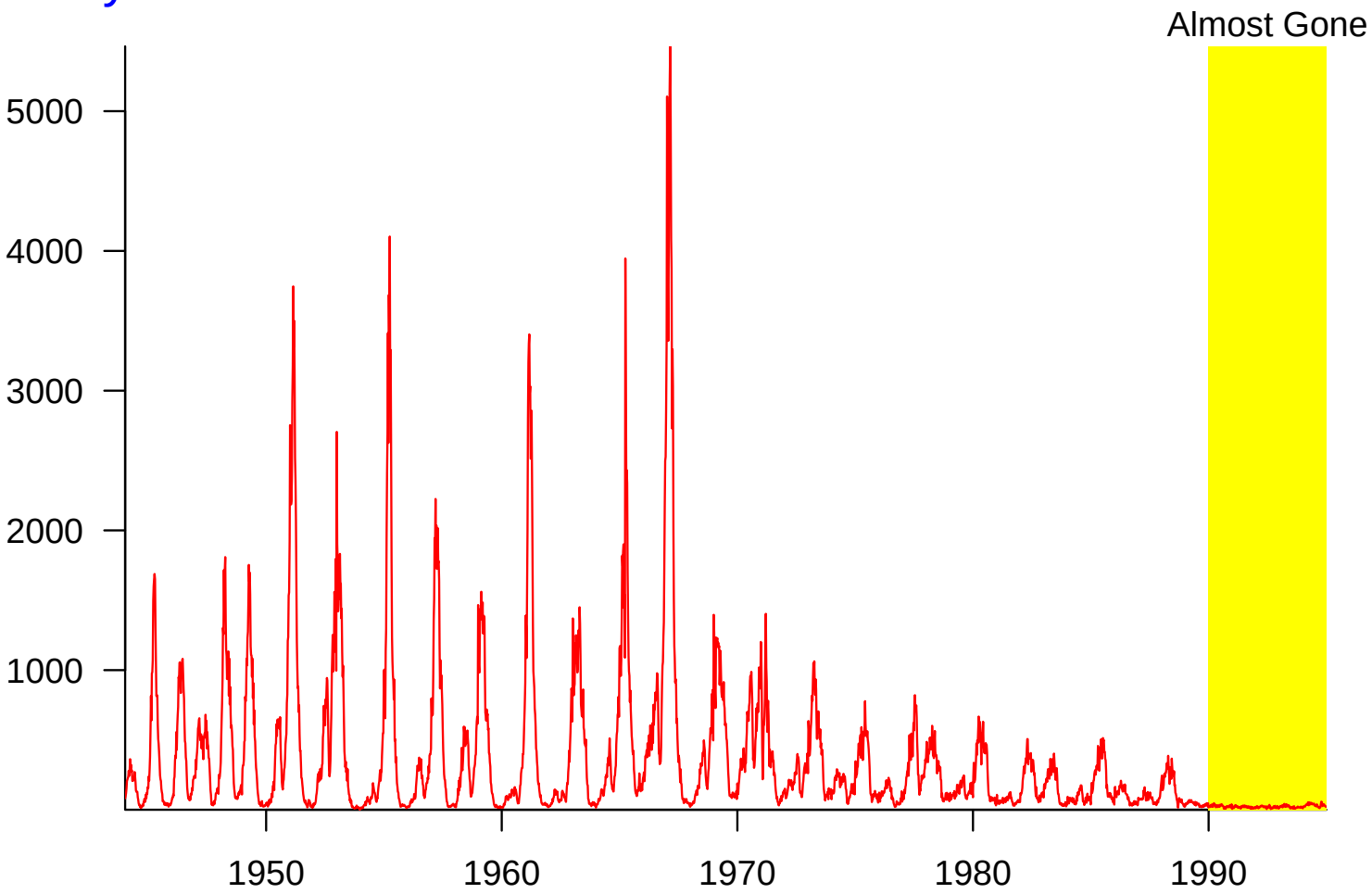
Weekly Cases



Earn, Rohani, Bolker, Grenfell, 2000, *Science* **287**, 667–670

Measles incidence in London, England, 1945–1995

Weekly Cases



Earn, Rohani, Bolker, Grenfell, 2000, *Science* **287**, 667–670

The seasonally forced SIR model

Many respiratory infections spread more easily in the winter. Consequently, the transmission rate β is ***seasonally forced***. A simple model of seasonal variation is a sinusoid:

$$\beta \quad \longrightarrow \quad \beta(t) = \beta_0(1 + \alpha \cos(2\pi t))$$

(t measured in years)

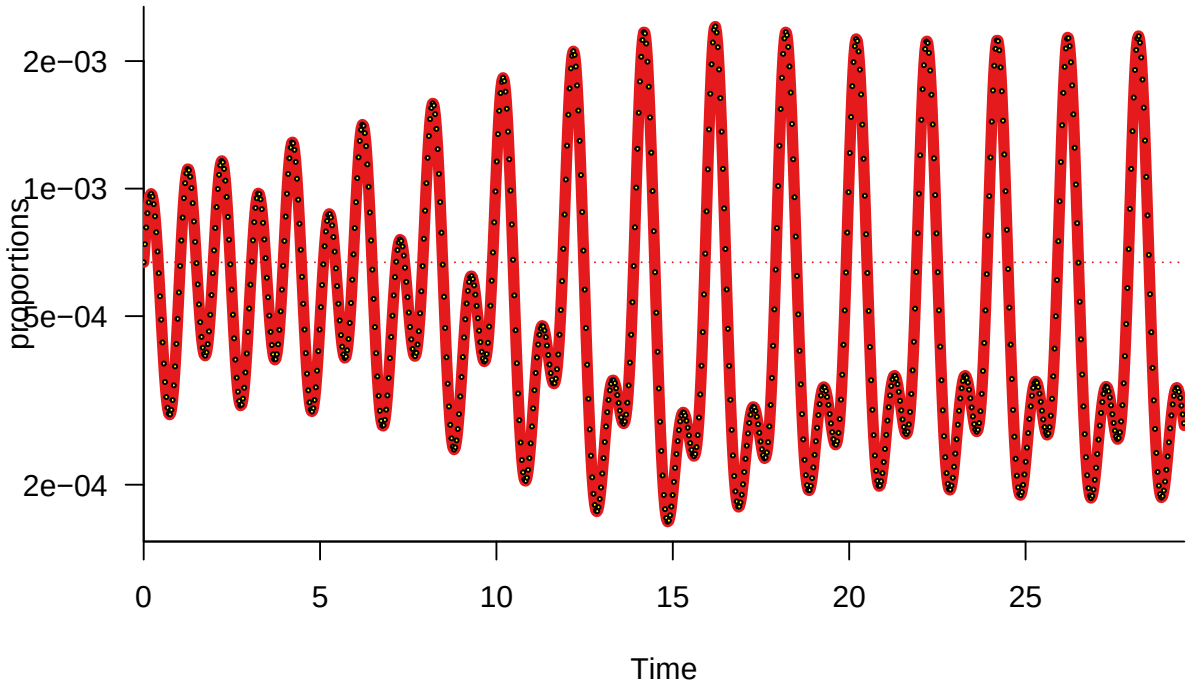
New parameter: $\alpha =$ ***amplitude of seasonal forcing***

Revised (non-autonomous) ***system of three forced ODEs***:

$$\frac{dS}{dt} = \mu - \beta(t)SI - \mu S, \qquad \frac{dI}{dt} = \beta(t)SI - \gamma I - \mu I, \qquad \frac{dR}{dt} = \gamma I - \mu R$$

The seasonally forced SIR model

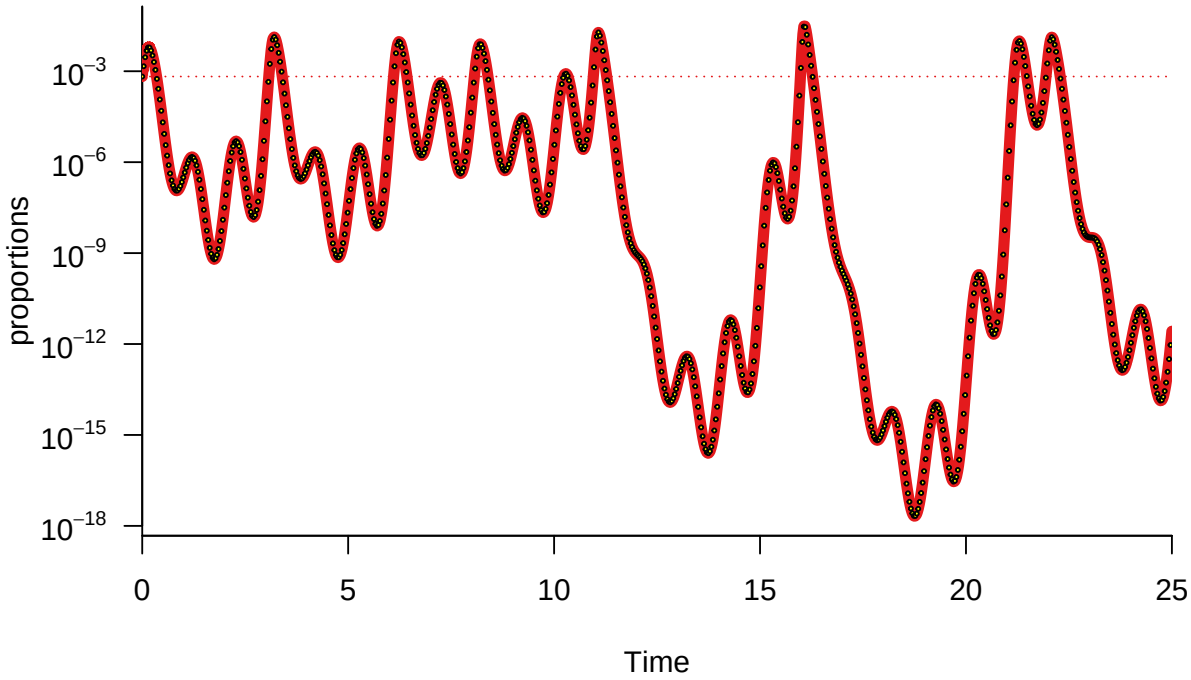
Oscillations (*epidemic cycles*) do not damp out when seasonally forced. There is no equilibrium with disease present. Asymptotic dynamics can be very complicated: *recurrent epidemics* with magnitudes that change *regularly*



Solution for $\mathcal{R}_0 = 17$, $\frac{1}{\gamma} = 13$ days, $\alpha = 0.1$

The seasonally forced SIR model

Oscillations (*epidemic cycles*) do not damp out when seasonally forced. There is no equilibrium with disease present. Asymptotic dynamics can be very complicated: *recurrent epidemics* with magnitudes that change *regularly* or can seem random (or "chaotic").



Solution for $\mathcal{R}_0 = 17$, $\frac{1}{\gamma} = 13$ days, $\alpha = 0.8$

Shameless plug for other courses

If nonlinear dynamics seems interesting, consider taking ***Math 3F03 – Advanced Differential Equations***.

If you take Math 3F03, you can follow that with ***Math 4MB3 – Mathematical Biology***.